WINTER INTERNSHIP 2026

GENERAL INTERNSHIP HANDBOOK

"Your Complete Guide to a Successful Internship"

Full Stack Development	Cybersecurity & Ethical Hacking	Digital Marketing

■ vectonix.in ■ 25 Nov – 25 Dec 2026 ■ 4 Weeks · Full-Time

Confidential · For Registered Interns Only · Vectonix Technologies Pvt. Ltd.

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CHAPTER 1

Welcome to Vectonix Technologies

Part 1 · Welcome & Introduction

Welcome Message

Dear Intern, welcome to the Vectonix Technologies Winter Internship 2026! We are genuinely thrilled to have you as part of our growing community of learners, builders, and future professionals. This internship is not just a line on your résumé — it is a structured, mentored, real-world experience designed to challenge you, grow you, and prepare you for a successful career in technology.

Over the next 25 days — from 25 November to 25 December 2026 — you will work through four carefully designed milestones in your chosen domain. You will submit real projects, receive personalised feedback, and ultimately earn a certificate that reflects your actual effort and achievement. We believe in learning by doing, and every task in this programme has been crafted with that philosophy at its core.

About Vectonix Technologies

Vectonix Technologies is a forward-thinking technology company with expertise across Full Stack Web Development, Cybersecurity & Ethical Hacking, and Digital Marketing. Founded with a mission to bridge the gap between classroom knowledge and industry expectations, Vectonix operates a flagship internship programme that has helped hundreds of students transform from learners into skilled professionals.

Our team consists of experienced practitioners who are passionate about mentorship. We do not just assign tasks — we explain the why, review the how, and celebrate the result. Visit us at **vectonix.in** to learn more about our services and community.

What Makes This Internship Unique

- Milestone-based structure: four progressive checkpoints keep you on track
- Domain specialisation: you go deep in one area instead of skimming many
- Mentor feedback: personalised reviews, not just automated scores
- Portfolio-ready output: projects are designed to showcase on GitHub and LinkedIn
- Certificate of completion: issued only on merit, so it carries real weight

Meet the Three Domains

Domain	What You Build	Core Skills
Full Stack Development	End-to-end web applications with frontend UI, backend APIs, and database integration	HTML/CSS, JavaScript, React, Node.js, Express, MongoDB

Domain	What You Build	Core Skills
Cybersecurity & Ethical Hacking	Security assessments, CTF challenges, penetration testing reports, and hardened systems	Linux, Networking, Kali tools, OS
Digital Marketing	Campaigns, SEO audits, social media strategy, analytics dashboards, and content calendars	Google Analytics, Meta Ads, SE

CHAPTER 2

Internship Overview

Objectives · Schedule · Expectations

Programme Objectives

Objective 1: Develop domain mastery

Each intern will build a deep, practical understanding of their chosen domain through hands-on projects and structured tasks, moving well beyond surface-level tutorials.

Objective 2: Build a professional portfolio

By the end of Week 4, every intern will have a polished, documented GitHub repository with real projects they can show to employers.

Objective 3: Cultivate professional habits

The programme teaches time management, documentation, communication, and code/work review practices used daily in industry teams.

Objective 4: Apply industry tools

Interns will get comfortable with the tools professionals actually use — VS Code, Git, GitHub, Figma, Kali Linux, Google Ads Manager, etc.

Objective 5: Earn a merit-based certificate

The Vectonix certificate is awarded only to interns who complete all four milestones with acceptable quality, making it a meaningful credential that employers recognise.

Duration and Schedule

Week	Dates	Milestone	Focus
Week 1	25 Nov – 01 Dec	Milestone 1	Planning & Fundamentals
Week 2	02 Dec – 08 Dec	Milestone 2	Core Implementation
Week 3	09 Dec – 15 Dec	Milestone 3	Testing & Improvement
Week 4	16 Dec – 25 Dec	Milestone 4	Final Project & Presentation

What You Will Achieve by the End

By 25 December 2026, a committed intern who follows this handbook will have: a live or deployable project in their domain, a structured GitHub repository with commits showing their entire journey, a presentation deck or report documenting their process and results, and a merit-based internship certificate from Vectonix Technologies. More importantly, you will have the confidence and muscle memory of someone who has shipped real work.

CHAPTER 3

Getting Started

Tools · Accounts · First-Week Checklist

How to Use This Handbook

This handbook is your single source of truth for the duration of the internship. Read it once cover-to-cover in Week 1, then refer back to specific sections as you reach each milestone. Templates in Part 3 are ready to copy; fill them in and submit them as directed by your mentor. Do not skip sections — even if a chapter seems unrelated to your domain, the professional practices described apply to everyone.

Required Tools by Domain

Tool / Platform	Full Stack Dev	Cybersecurity	Digital Marketing
VS Code (IDE)	✔ Required	✔ Required	Optional
Git & GitHub	✔ Required	✔ Required	✔ Required
Browser DevTools	✔ Required	✔ Required	✔ Required
Node.js & npm	✔ Required	—	—
Kali Linux / Parrot OS	—	✔ Required	—
Burp Suite / OWASP ZAP	—	✔ Required	—
Wireshark	—	Recommended	—
Google Analytics / Search Console	Optional	—	✔ Required
Meta Business Suite	—	—	✔ Required
Canva / Figma	Optional	—	✔ Required
Postman	✔ Required	Recommended	—
MongoDB Compass / pgAdmin	Recommended	—	—

Creating Your Accounts

- GitHub — Create a free account at github.com. Use a professional username (e.g., firstname-lastname). Enable two-factor authentication.
- Google Account — Required for Google Drive (project submissions), Google Analytics (DM), and Google Classroom (if used by your mentor).

- LinkedIn — Set up or update your profile. You will add your internship here at the end of Week 4.
- Discord / Slack — Your mentor will share an invite link for the team workspace. Join within 24 hours of receiving it.
- Domain-specific tools — FSD interns: install Node 20 LTS and MongoDB Community. Cybersec interns: set up a VM with Kali Linux 2024.x. DM interns: create a free Google Analytics property.

First-Week Checklist

#	Task	Status
1	Read this handbook cover-to-cover	■ To Do
2	Set up GitHub account and create internship repository	■ To Do
3	Install all required tools for your domain	■ To Do
4	Join the team communication workspace (Discord/Slack)	■ To Do
5	Introduce yourself in the #introductions channel	■ To Do
6	Submit Day 1 Work Log (template in Part 3)	■ To Do
7	Attend orientation call or watch recording	■ To Do
8	Read Milestone 1 brief for your domain	■ To Do
9	Complete domain-specific environment setup	■ To Do
10	Submit first Weekly Progress Report by end of Week 1	■ To Do

CHAPTER 4

Professional Conduct

Code of Conduct · Communication · Escalation

Internship Code of Conduct

As a Vectonix intern, you represent the company and your own professional brand. The following ten rules are non-negotiable. Violations may result in disqualification from the certificate or removal from the programme.

Rule 1: Honesty and academic integrity

All submitted work must be your own. Using AI-generated code or copy-pasting solutions without understanding them is considered academic dishonesty and will result in milestone failure.

Rule 2: Respect and inclusivity

Treat all interns, mentors, and staff with respect regardless of background, skill level, or domain. Harassment, discrimination, or hostile communication of any kind is grounds for immediate removal.

Rule 3: Punctuality and deadlines

Milestone deadlines are firm. A 24-hour extension may be requested at least 48 hours in advance with a valid reason. Missing two consecutive deadlines without notice constitutes withdrawal from the programme.

Rule 4: Communication responsiveness

Respond to mentor messages within 24 hours on working days. If you cannot, notify your mentor in advance. Disappearing without notice for more than 3 days will trigger a review.

Rule 5: Quality over quantity

Submit work that you are genuinely proud of, not just work that fills the requirement. Mentors are evaluating depth of understanding and quality of execution, not word count or line count.

Rule 6: Confidentiality

Do not share proprietary materials, mentor feedback, internal Slack/Discord content, or project briefs outside the internship group without written permission.

Rule 7: Professional language

All communication — written or verbal — must remain professional. Slang, offensive language, and informal complaints in public channels are not appropriate.

Rule 8: Attribution and credits

If you use open-source libraries, tutorials, or external resources, cite them clearly in your README and project documentation. Uncited copying is plagiarism.

Rule 9: Engagement and participation

Actively participate in group sessions, feedback calls, and community activities. Passive participation (only submitting tasks without engaging) limits your learning and affects your final evaluation.

Rule 10: Device and tool responsibility

You are responsible for your own hardware, internet connectivity, and development environment. Plan for outages. Back up your work to GitHub daily.

How to Reach Your Mentors

Channel	Best For	Response Time
Slack / Discord DM	Quick questions, status updates, clarifications	Within 24 hours
Email (mentor address)	Formal requests, extension applications, complaints	48 hours
Scheduled video call	In-depth feedback, milestone reviews, stuck > 2 hours	Booked in advance
Community help channel	Questions the whole group can benefit from	24 hours (by anyone)

What to Do When You Are Stuck

1. Spend 20–30 minutes trying to solve it yourself. Googling the error message, reading docs, and re-reading your code are all valid first steps.
2. Check if the problem is already solved in the community help channel — search before asking.
3. Post a detailed question in the help channel: describe what you tried, share the error message, and show relevant code.
4. If unsolved after 2 hours of genuine effort, DM your mentor directly with the same information.
5. Never stay stuck silently for more than one day. Asking for help is a professional skill, not a weakness.

CHAPTER 5

Project Evaluation System

Milestones · Grading · Certificate Criteria

How Projects Are Evaluated

Evaluation at Vectonix is holistic and milestone-based. Each of the four milestones is scored independently, and your overall internship grade is the weighted average of all four scores. A minimum score of 60% across all milestones is required for certificate issuance. Scores below 60% in any single milestone will trigger a mentor review and an opportunity to revise within 48 hours.

Grading Criteria

Criterion	Weight	What Mentors Look For
Technical Correctness	35%	Does the solution work as specified? Are there bugs or broken features
Code / Work Quality	25%	Is the code clean, documented, and following best practices? Is the deli
Documentation	20%	README, comments, work logs, and reports are clear, complete, and a
Creativity & Initiative	10%	Did the intern go beyond the minimum? Did they explore, experiment, o
Timeliness	10%	Was the milestone submitted on time? Were daily logs maintained?

Certificate Issuance Criteria

Score Range	Certificate Grade	Status
90 – 100%	Distinction	Issued with highest honour
75 – 89%	Merit	Issued with merit notation
60 – 74%	Pass	Standard certificate issued
Below 60%	Fail	Certificate withheld; revision window offered

MILESTONE 1

Planning & Fundamentals

Week 1 · 25 Nov – 01 Dec 2026

Milestone Objective: Establish your development environment, understand the project brief, and complete all foundational tasks that prove readiness for Week 2 implementation.

Full Stack Development	Cybersecurity & Ethical Hacking	Digital Marketing
• Set up Git repo, VS Code, Node.js, and database (MongoDB or PostgreSQL) • Create a wireframe/mockup of your final project UI (Figma or hand-drawn scanned) • Build and deploy a simple static HTML/CSS personal page as a warm-up task • Write a Project Plan document outlining your Week 2–4 development roadmap • Complete the JavaScript Fundamentals assessment (linked in Discord)	• Install and configure Kali Linux in a virtual machine (VirtualBox / VMware) • Run your first network scan using Nmap on a provided CTF environment • Document five OWASP Top 10 vulnerabilities with a real-world example each • Complete a basic web challenge on TryHackMe or HackTheBox Starter • Set up Burp Suite Community and capture your first HTTP request	• Create a Google Analytics 4 property for a provided test website • Conduct a 10-point SEO audit on an assigned website using free tools • Build a 30-day social media content calendar for a fictional brand • Write three blog posts (600+ words each) optimised for a target keyword • Deliver a written competitor analysis of three brands in your assigned niche

Expected Deliverables

- GitHub repository initialised with README and project structure
- Milestone 1 Report (use template in Part 3) submitted via Google Drive
- Day 1–7 Work Logs completed and submitted
- First Weekly Progress Report submitted

MILESTONE 2

Core Implementation

Week 2 · 02 Dec – 08 Dec 2026

Milestone Objective: Build the primary deliverable for your domain. This week's work forms the core of your final project. Quality and completeness matter more than speed.

Full Stack Development	Cybersecurity & Ethical Hacking	Digital Marketing
• Implement the backend API with at least 5 RESTful endpoints (CRUD for one resource) • Build the frontend UI with React or plain JS — connect it to your backend • Set up user authentication (JWT or session-based) with register and login routes • Write Postman collection documenting all API endpoints with sample requests/responses • Push all code to GitHub with meaningful commit messages after each feature	• Perform a full reconnaissance phase on the assigned lab target (passive + active) • Exploit one identified vulnerability on the lab environment and document the process • Write a professional Penetration Testing Report (template in Part 3) • Complete two intermediate-level rooms on TryHackMe in your specialisation path • Configure and run OWASP ZAP against a provided test web application	• Launch and manage a low-budget Google Ads or Meta Ads test campaign (with provided credits) • Build a keyword research document with 50+ ranked keywords for your niche • Publish three SEO-optimised blog posts on the provided WordPress test site • Create a campaign performance tracking spreadsheet with key KPI columns • Design three social media post graphics using Canva and schedule them with Buffer

Expected Deliverables

- Core project pushed to GitHub with clean commit history
- Milestone 2 Report with screenshots and explanations
- Day 8–14 Work Logs
- Second Weekly Progress Report

MILESTONE 3

Testing & Improvement

Week 3 · 09 Dec – 15 Dec 2026

Milestone Objective: Test your Week 2 work rigorously, address bugs and weaknesses, and add the polish that separates functional from professional. Incorporate mentor feedback from Milestone 2.

Full Stack Development	Cybersecurity & Ethical Hacking	Digital Marketing
• Write unit tests for at least three backend functions using Jest or Mocha • Perform manual user testing with three testers and document issues found • Fix all critical bugs identified in Milestone 2 feedback • Improve UI responsiveness — test on mobile, tablet, and desktop widths • Optimise one database query and document the performance improvement	• Enumerate and exploit a second vulnerability on the lab — use a different attack vector • Document your methodology using the PTES (Penetration Testing Execution Standard) format • Harden the test system against your identified vulnerabilities and verify with a rescan • Write a recommendations section in your pen test report with CVSS severity scores • Complete the Privilege Escalation room on TryHackMe	• Analyse Week 2 campaign data — identify three underperforming ads and optimise them • Write an A/B test plan for two landing page variants and implement basic changes • Produce a mid-campaign analytics report with GA4 screenshots and interpretation • Review and improve on-page SEO for all three published blog posts • Create a detailed email marketing sequence (5 emails) with subject lines and body copy

Expected Deliverables

- Updated GitHub repo reflecting all fixes and improvements
- Milestone 3 Report including test results and mentor-feedback responses
- Day 15–21 Work Logs
- Third Weekly Progress Report

MILESTONE 4

Final Project & Presentation

Week 4 · 16 Dec – 25 Dec 2026

Milestone Objective: Complete your final project, prepare a professional presentation, write the Final Project Report, and deliver an internship presentation to your mentor panel.

Full Stack Development	Cybersecurity & Ethical Hacking	Digital Marketing
• Deploy your full-stack application to a live hosting platform (Render, Vercel, Railway, etc.) • Record a 5-minute demo video walking through all features • Write the Final Project Report (template in Part 3) with architecture diagram • Prepare and deliver a 10-minute live or recorded presentation to your mentor • Update GitHub README with live demo link, screenshots, and setup instructions	• Complete and finalise your Penetration Testing Report with executive summary • Create a risk matrix mapping all vulnerabilities to business impact • Prepare a 10-minute debrief presentation as if presenting to a client • Write a personal reflection on what you would do differently in a real engagement • Upload all lab reports, tool outputs, and notes to GitHub in a structured folder	• Prepare a final campaign performance report with all KPIs from Weeks 2–3 • Build a comprehensive digital marketing strategy document for a provided brand brief • Create a 10-minute presentation summarising campaign results and future recommendations • Deliver your presentation to your mentor and respond to Q&A; • Compile your full internship portfolio into a PDF (all reports, posts, calendars, graphics)

Expected Deliverables

- Final GitHub repository — clean, documented, complete
- Final Project Report (PDF)
- Presentation recording or live session with mentor
- All four Weekly Progress Reports and all 25 daily Work Logs
- Certificate application form submitted (link shared in Discord on Dec 20)

PART 3

Common Resources

Templates · Guides · Rubrics

Weekly Progress Report Template

Submit this report at the end of every week via the shared Google Drive folder your mentor will create. The sample below shows Week 1 filled in — use it as a guide for your own submissions.

Field	Sample (Week 1 — Full Stack Development)
Intern Name	Arjun Mehta
Week Number	Week 1 (25 Nov – 01 Dec 2026)
Domain	Full Stack Development
Mentor	Ms Priya Sharma
Tasks Completed	Env setup ✔ | Wireframe ✔ | Static page deployed ✔ | Project Plan ✔ | JS asses
Tasks Pending / Incomplete	None — all Milestone 1 tasks completed
Key Learning This Week	Understood the difference between synchronous and asynchronous JS. Learned how
Blockers or Challenges	Spent 3 hours debugging a CORS error — resolved by adding the cors middleware.
Plan for Next Week	Start building the backend API. Implement User model and auth routes first, then mov
GitHub Commits This Week	14 commits across 3 branches
Daily Log Submitted?	All 7 days ✔
Self-Rating (1–10)	8 — strong week, could have started one day earlier

Daily Work Log Template

Field	Sample Entry (Day 3 — 27 Nov 2026)
Date	27 November 2026
Domain & Intern	Full Stack Dev · Arjun Mehta
Work Hours Today	6 hours (10:00 AM – 5:00 PM, 1 hr break)
Morning Goal	Complete the Figma wireframe for the homepage and product listing page

Field	Sample Entry (Day 3 — 27 Nov 2026)
Tasks Done	Finished homepage wireframe. Created component map (Header, Hero, ProductGrid,
Tasks Not Done	Product detail page wireframe — moving to Day 4
Learnings	Learned Figma's Auto Layout feature — speeds up responsive design mockups signif
Blockers	None today
Tomorrow's Plan	Finish product detail wireframe. Write the Project Plan document — target 800 words.
GitHub Activity	2 commits: added wireframe screenshots to /docs folder
Mood / Energy	Good — productive day

GitHub Submission Guide

All project work must be submitted via GitHub. Your repository is your portfolio — treat it with care. Follow these steps exactly.

Step 1: Create your internship repository

Go to github.com, click "New repository". Name it: vectonix-internship-2026. Set it to Public. Initialise with README. Click "Create repository".

Step 2: Clone the repository locally

Open Terminal or Git Bash. Run:

```
git clone https://github.com/YOUR_USERNAME/vectonix-internship-2026
```

```
cd vectonix-internship-2026
```

Step 3: Set up your folder structure

Create folders: /milestone-1 /milestone-2 /milestone-3 /milestone-4 /docs /reports

Add a .gitignore file (use the Node or Python template from github.com/github/gitignore).

Step 4: Work in branches

Never commit directly to main for project work. Create a branch per milestone:

```
git checkout -b milestone-1
```

Work, commit, push, then open a Pull Request to main.

Step 5: Write good commit messages

Format: :

Examples:

```
feat: add user registration API endpoint
```

```
fix: resolve CORS error in auth middleware
```

```
docs: update README with deployment link
```

Step 6: Submit milestone link

After merging your PR to main, copy the repository URL and paste it into the Milestone Submission Form (link pinned in Discord). Include the specific folder path for the milestone if requested.

Presentation Preparation Guide

Slide #	Title	Content to Include
1	Cover Slide	Your name, domain, internship title, date
2	About Me	Background, goals, why you chose this domain
3	Project Overview	What you built/did, the problem it solves, target audience
4–5	Technical Deep Dive	Architecture diagram, key features, tools used, code snippets or screenshots

Slide #	Title	Content to Include
6	Challenges & Solutions	Two or three specific challenges you faced and exactly how you resolved them
7	Demo / Results	Live demo or screenshots; key metrics (FSD: features; Cyber: vulnerabilities found; I
8	Key Learnings	Top three technical skills gained; one soft skill improved
9	What I'd Do Differently	Honest reflection — shows maturity and self-awareness
10	Q&A	Thank the panel; invite questions; have a one-sentence project summary ready

Final Project Evaluation Rubric

Criterion	Marks	Distinction (90–100)	Pass (60–74)
Technical Implementation	35	Fully functional, deployed, zero critical bugs, clean architecture	Functional with minor bugs, not d
Code / Work Quality	25	Excellent structure, comments, naming, follows best practices throughout	Readable with some issues; inco
Documentation	20	Comprehensive README, all reports submitted, architecture diagram present	Basic README, most reports su
Creativity & Initiative	10	Went beyond brief: extra features, experiments, or unique approach	Met the brief exactly, minimal ex
Timeliness	10	All four milestones and daily logs submitted on time	One or two late submissions with
TOTAL	100	90+ = Distinction Certificate	60–74 = Pass Certificate

PART 4

Knowledge Assessment

MCQs · Short Answers · Practical Tasks

This section tests your understanding of the internship programme, professional practices, and domain fundamentals. Complete this assessment in Week 1 and submit your answers via the Google Form your mentor will share. Answers are provided at the end of this section for self-review after submission.

Section A — Multiple Choice Questions (5 × 4 marks = 20)

1. What is the minimum average score required across all four milestones to receive an internship certificate?

A. 50%

B. 55%

C. 60%

D. 70%

Answer: C. 60% — As stated in Chapter 5, a minimum of 60% across all milestones is required for certificate issuance.

2. Which of the following is NOT an accepted reason to request a deadline extension?

A. Medical emergency with documentation

B. You forgot about the deadline

C. Family emergency communicated 48 hours in advance

D. Technical failure affecting your work environment

Answer: B. You forgot about the deadline — Forgetting a deadline is not a valid reason for extension. All other options represent circumstances beyond the intern's control.

3. When should you push your internship work to GitHub?

A. Only at the end of each milestone

B. Once per week on Sundays

C. Daily, after each meaningful piece of work

D. Only when your mentor asks

Answer: C. Daily, after each meaningful piece of work — Chapter 3 (First-Week Checklist) and Chapter 4 (Code of Conduct) both emphasise daily GitHub backups.

4. What does the 'Creativity & Initiative' criterion (10%) reward?

A. Using the most popular frameworks

B. Submitting the most files

C. Going beyond the minimum brief with exploration or extras

D. Having the longest commit history

Answer: C. Going beyond the minimum brief with exploration or extras — The rubric defines this criterion as rewarding interns who go beyond the brief, experiment, and add unique value.

5. Which tool would a Cybersecurity intern use to capture and inspect HTTP traffic during a web application assessment?

A. VS Code

B. Google Analytics

C. Canva

D. Burp Suite

Answer: D. Burp Suite — Burp Suite is a web security testing proxy used by cybersecurity professionals to intercept, inspect, and modify HTTP/S traffic.

Section B — Short Answer Questions (5 × 8 marks = 40)

1. Describe in your own words what 'academic integrity' means in the context of this internship. Give one example of a violation and explain its consequence.

Answer: _____

2. Your application's login route is returning a 500 Internal Server Error. Walk through the exact steps you would take to diagnose and fix the issue.

Answer: _____

3. Why is it important to write meaningful Git commit messages? Write three example commit messages for tasks you might do in Week 2.

Answer: _____

4. A digital marketing intern notices their Google Ads campaign has a high click-through rate (CTR) but a very low conversion rate. What could be causing this, and what would you investigate first?

Answer: _____

5. You are a cybersecurity intern who has discovered a critical SQL injection vulnerability on a test application. Describe the three most important steps in your reporting process.

Answer: _____

Section C — Practical Tasks (2 × 20 marks = 40)

Practical Task 1 (All Domains)

Create a GitHub repository called 'vectonix-practice'. Inside it, create a file called 'REFLECTION.md'. In that file, write 200–300 words describing: (a) your domain choice and why you made it, (b) one technical skill you want to master by the end of the internship, and (c) how you plan to stay accountable. Push the file and share the repository link with your mentor.

Practical Task 2 (Domain-Specific)

FSD: Build a single-page todo app with JavaScript — no frameworks. It must add, complete, and delete tasks. Deploy it to GitHub Pages. Cybersecurity: Using a free account on TryHackMe, complete the 'Introduction to Web Hacking' room. Screenshot each completed task and compile them into a PDF report. Digital Marketing: Write an 800-word SEO article on the topic 'Top 5 Free Tools for Small Business Digital Marketing'. Use one target keyword in the title, first paragraph, one heading, and the last paragraph.

PART 5

Glossary of Common Terms

20+ Terms from All Three Domains

The following terms appear throughout this handbook and across your domain tasks. Understanding them precisely will help you communicate professionally with your mentor and colleagues. Terms are arranged alphabetically.

Term	Simple Meaning	Example
API (Application Programming Interface)	A defined set of rules that allows two software systems to communicate with each other.	*A weather app calls a weather API to get current temperature data.*
Attack Vector	The path or method an attacker uses to gain unauthorised access to a system.	*A phishing email is an attack vector that delivers malware via a link.*
Backlink	A link from another website pointing to your site — a key factor in SEO ranking.	*A tech blog linking to your article on 'JavaScript tips' is a backlink.*
Bounce Rate	The percentage of website visitors who leave after viewing only one page.	*A landing page with 80% bounce rate means most visitors left without clicking further.*
CRUD	Create, Read, Update, Delete — the four basic operations on any data store.	*A blog app does CRUD on posts: write a post, read it, edit it, delete it.*
CTF (Capture The Flag)	A cybersecurity competition where participants solve security challenges to find hidden flags.	*TryHackMe and HackTheBox host CTF rooms for skill-building.*
CTR (Click-Through Rate)	The ratio of users who click an ad or link to the total number who saw it.	*1,000 impressions and 50 clicks = 5% CTR.*
CVSS (Common Vulnerability Scoring System)	A standardised framework for rating the severity of security vulnerabilities on a 0–10 scale.	*A CVSS score of 9.8 is 'Critical'; a score of 2.0 is 'Low'.*
DOM (Document Object Model)	The browser's in-memory tree representation of an HTML page, which JavaScript can manipulate.	*document.getElementById('title').textContent = 'Hello' changes the DOM.*
Endpoint	A specific URL in an API where a client can send a request to trigger a server action.	*POST /api/users/register is an endpoint that creates a new user account.*

Enumeration	The process of systematically gathering information about a target system, network, or user.	*Running Nmap to discover open ports is a form of network enumeration.*
Git	A distributed version control system that tracks changes to files over time.	*git commit -m 'add login feature' saves a snapshot of your current code.*
JWT (JSON Web Token)	A compact, URL-safe token used to securely transmit identity information between parties.	*After login, the server returns a JWT; the client sends it with every protected request.*
KPI (Key Performance Indicator)	A measurable value that tracks progress toward a specific business objective.	*For a DM campaign, KPIs might be CTR, Cost per Lead, and Conversion Rate.*
Middleware	A function in a web framework that runs between receiving a request and sending a response.	*An authentication middleware checks if a user is logged in before allowing access to a route.*
OWASP Top 10	A standard awareness document listing the ten most critical web application security risks.	*SQL Injection and Cross-Site Scripting (XSS) appear in every edition of the OWASP Top 10.*
Payload	The actual data content sent in a request or, in security, the malicious code in an exploit.	*A POST request's JSON body is the payload. In an XSS attack, a script tag is the payload.*
REST (Representational State Transfer)	An architectural style for building APIs using standard HTTP methods (GET, POST, PUT, DELETE).	*A RESTful API for a bookshop uses GET /books to list books and POST /books to add one.*
ROAS (Return on Ad Spend)	Revenue generated for every currency unit spent on advertising.	*Spending ■1,000 and earning ■4,000 in sales = 4x ROAS.*
SEO (Search Engine Optimisation)	The practice of optimising content and websites to rank higher in organic search results.	*Using the keyword 'best free antivirus' in your H1, meta description, and article body is on-page SEO.*
SQL Injection	A vulnerability where an attacker inserts malicious SQL code into an input to manipulate a database.	*Entering ' OR '1'='1 in a login field can bypass authentication on a vulnerable site.*
Version Control	A system that records changes to files so you can recall specific versions later.	*Git is the most widely used version control system in software development.*
VM (Virtual Machine)	Software that emulates a complete computer, allowing you to run a different OS inside your current one.	*A cybersecurity intern runs Kali Linux inside VirtualBox — a VM on their Windows laptop.*

Wireframe	A low-fidelity visual guide showing the layout of a web page without final design details.	*A wireframe for a login page shows boxes for email, password, and a submit button — no colours.*

We wish you a productive, rewarding, and career-defining internship. The effort you put in over the next 25 days will compound for years to come. Give it everything you have — we will be with you every step of the way.

— The Vectonix Technologies Mentorship Team

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